

Lab 8 Report

Ryan Wilson

Due: 03/31/2026 @ 11:00 AM

1 Integer Ranges

Table 1: Integer ranges by size				
Size	Unsigned Min. Value	Max. Value	Signed Min. Value	Max. Value
8-bit	0	255	-128	127
16-bit	0	65,535	-32,768	32,767
32-bit	0	4,294,967,295	-2,147,483,648	2,147,483,647
64-bit	0	18,446,744,073,709,551,615	-9,223,372,036,854,775,808	9,223,372,036,854,775,807

2 8-bit Unsigned Representations

Decimal	Binary
98	01100010
0	00000000
1	00000001
127	01111111
255	11111111

3 8-bit 2's Complement Signed Representations

Decimal	Binary
+98	01100010
-44	11010100
-1	11111111
0	00000000
+1	00000001
-128	10000000
+127	01111111

4 32-bit Floating Points (Normalized and Denormalized)

- **Normalized**

- Smallest positive: $1.0 \times 2^{-126} \approx 1.175 \times 10^{-38}$
- Largest positive: $(2 - 2^{-23}) \times 2^{127} \approx 3.403 \times 10^{38}$
- Smallest negative: $-(2 - 2^{-23}) \times 2^{127} \approx -3.403 \times 10^{38}$
- Largest negative: $-1.0 \times 2^{-126} \approx -1.175 \times 10^{-38}$

- **Denormalized**

- Smallest positive: $2^{-149} \approx 1.4 \times 10^{-45}$
- Largest positive: $(1 - 2^{-23}) \times 2^{-126} \approx 1.175 \times 10^{-38}$
- Smallest negative: $-(1 - 2^{-23}) \times 2^{-126} \approx -1.175 \times 10^{-38}$
- Largest negative: $-2^{-149} \approx -1.4 \times 10^{-45}$

5 64-bit Floating Points (Normalized and Denormalized)

- **Normalized**

- Smallest positive: $2^{-1022} \approx 2.225 \times 10^{-308}$
- Largest positive: $(2 - 2^{-52}) \times 2^{1023} \approx 1.797 \times 10^{308}$
- Smallest negative: $-(2 - 2^{-52}) \times 2^{1023} \approx -1.797 \times 10^{308}$
- Largest negative: $-2^{-1022} \approx -2.225 \times 10^{-308}$

- **Denormalized**

- Smallest positive: $2^{-1074} \approx 4.9 \times 10^{-324}$
- Largest positive: $(1 - 2^{-52}) \times 2^{-1022} \approx 2.225 \times 10^{-308}$
- Smallest negative: $-(1 - 2^{-52}) \times 2^{-1022} \approx -2.225 \times 10^{-308}$
- Largest negative: $-2^{-1074} \approx -4.9 \times 10^{-324}$